3. Polishing lathe is a type of
   a. Engine lathe
   b. Tool room lathe
   c. High speed lathe
   d. Automatic lathe
4. The spindle speed of high speed lathe is around
   a. 600-1000 rpm
   b. 1200-3200 rpm
   c. 2000-2400 rpm
   d. 3500-4000 rpm



5. A part used in taper turning is
   a. Cross-slide
   b. Compound rest
   c. Saddle
   d. Apron
6. The formula for finding the cutting speed in a lathe is

   a. $\frac{\pi DN}{1000}$ b. $\frac{d_1 - d_2}{2l}$

   c. $\pi r^2 h$ d. $\frac{D-d}{2} \times \frac{L}{l}$

Part – II

Answer the following questions in one or two sentences: **(3 marks)**

7. What is turning?
8. What are the uses of tool room lathe?
9. Mention the special features of turret and capstan lathe.
10. What is swing diameter?
11. What are the special features of V-bed in a lathe?
12. What are the uses of back gear in a lathe?
13. Define 'Cutting speed' in a lathe.
14. What is the use of Lead screw in a lathe?
15. Mention the use of a face-plate.
16. Mention the formula to use the cutting of a thread, in a lathe.

Part – III

Answer the following questions in about a page **(5 marks)**

17. List out the types of lathe.
18. Mention any five differences between a turret and a capstan lathe.
19. What are the parts in the carriage of a lathe? Explain any one part.
20. Mention the different specifications of a lathe.
21. Sketch and explain the taper-turning by using the form-tool method.

Part – IV

Answer the following questions in detail **(10 Marks)**

22. Draw a neat sketch of a lathe and label its parts.
23. Explain with neat sketch of a step-cone pulley drive mechanism.
24. Explain any two operations performed in lathe with neat sketch.
25. Explain any two work holding devices in a lathe with suitable diagrams.

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